

Case Study

Clearview Properties

You have just joined the analytics team of a real estate advisory firm that helps clients buy, sell, and price residential properties in a mid-sized U.S. city. Until now, the firm has relied on the intuition of senior agents and a handful of recent comparable sales to estimate prices. Leadership suspects there is room for improvement and wants to start using their historical data more seriously.

About the client

Clearview Properties is a real estate agency operating in Cedar Falls, Iowa (United States). They work with both homeowners who want to sell and buyers looking for a home. Their commercial team is made up of brokers who negotiate directly with clients.

In recent weeks, the directors have identified two situations creating friction in their day-to-day work:

- › When a homeowner asks for an initial valuation to list their property, brokers take a long time to provide a reasonable range. Each broker estimates “by feeling” based on their experience, and prices are not consistent across the team.
- › When a homeowner asks “what should I improve in my house to sell it at a better price?”, the answer is also subjective. They have no clear evidence about which characteristics of a home weigh more on the final price, nor which investments are actually worth making.

The client is reaching out to us because they want to back up their broker team with concrete evidence. They are not asking for “a magic tool”. They want to understand their own market.

What the client wants the analysis to answer

The specific questions they need to answer are:

- › **Q1:** Given a property with certain characteristics, what is a reasonable price range to list it at? They need a reference that brokers can use as a starting point in the conversation with the owner.
- › **Q2:** Which characteristics of a property have the largest impact on its sale price? Ideally, split between characteristics that the owner can modify (kitchen quality, renovations, garage condition) and those that are structural or location-based (lot, neighborhood, year built).
- › **Q3:** Among the characteristics that the owner can modify, which ones are a profitable investment? In other words, which improvements are worth making relative to their cost, and which generate the best return in sale price?

- › **Q4:** Is there a clear pattern or market segmentation worth knowing about? For example, segments of properties that behave differently from the rest.

Available data

The client provides files containing a historical record of property transactions in Cedar Falls, Iowa, over several years. Each row represents a sold property, including information about the property itself, its location, the sale date, and the final price.

File	Description
Data.csv	Historical transactions. One row per sold property. Approximately 2,960 transactions with ~80 variables.
Data_Dictionary.txt	Description of each variable: meaning, possible values, units. Essential to correctly interpret the coded categorical columns.

Important: the data comes as the client handed it over. It is the technical team's responsibility to review the quality of the information before producing any conclusions. The client may not be aware of problems in their own dataset, and it is up to the consultant to identify, document and communicate them.

Expected deliverables

1 • Technical notebook (.ipynb)

A single notebook that documents the whole process end-to-end: how the data was explored, what was found, what decisions were made and why. The code must be reproducible; someone on the technical team should be able to run it and get the same results.

At a general level, the notebook should include the full data science workflow: data analysis, modeling, and numerical justifications supporting each decision and conclusion. We do not prescribe the steps, the structure, the techniques or the order; those choices are part of what is being evaluated. What we expect to see is that any claim, decision or recommendation is backed by evidence visible in the notebook (numbers, charts, metrics) rather than asserted.

2 • Client-facing presentation (.pptx or .pdf)

A presentation addressed to the general director of Clearview Properties, who has no technical background. The presentation must answer the client's questions using business language, not data science language. Aim for 8–12 slides.

What the client expects to see:

- › A clear summary of what was found in response to each of their questions.
- › Actionable recommendations that the brokers can start using next week.
- › Visualizations that a non-technical person can read without prior explanation.
- › An honest mention of assumptions, limitations and suggested next steps.

We do not expect the presentation to include code, detailed statistical metrics, or terms such as R^2 , MAE, regularization, etc. If at any point something like that needs to be mentioned, it must be translated into everyday language.

How we evaluate the work

The goal of this assessment is not to arrive at “the correct number”. We evaluate the rigor of the analytical process, the soundness of the methodological decisions, and the numerical evidence behind them, and the ability to translate the technical work into useful answers for a non-technical client.

Time, format and delivery

Recommended time: 4 calendar days from receiving this brief. Approximately 6 hours of work.

How to deliver

- › Technical notebook in .ipynb (Jupyter) format.
- › Client-facing presentation in .pptx or .pdf format.
- › Any intermediate file you consider relevant (cleaned datasets, scripts, etc.).
- › Everything delivered in a single .zip archive or via a Git repository (public, or private with shared access). If delivering as a zip, include a short note on how to run the notebook (dependencies, Python version, or any setup step needed).

Follow – up

A conversation of about 45 minutes will be scheduled after submission, with 20 minutes dedicated to the case study.

Come prepared to walk us through your reasoning and to defend your choices.

Considerations

- › Any Python or R library is allowed, as well as any visualisation tool.
- › If you decide to use auxiliary tools (AI chatbots, templates, etc.), we ask that you declare it in the notebook. We are evaluating how you reason, not whether you did it alone.
- › It is not necessary to reach “the perfect answer”. An honest analysis, well documented, with clear limitations is preferable to a polished result without justification.

Any questions during the assessment can be directed to the recruitment team at smoreno@cuestapartners.com.